

CRANBROOK
SCHOOL

HSC Trial Examination

Mathematics Extension 2

Friday July 23, 2010

General Instructions

- Reading time – 5 minutes
- Writing time – 3 hours
- All 8 questions should be attempted
- Total marks available - 120
- All questions are worth 15 marks
- Begin a new 8 page booklet for each question
- An approved calculator may be used
- A table of standard integrals can be found at the back of the paper
- All relevant working should be shown for each question

Question 1 (15 marks) Use a new 8 page booklet

Marks

(a) Find : $\int x\sqrt{3x-1} \, dx$ 3

(b) By using the substitution $t = \tan \frac{\theta}{2}$, evaluate

$$\int_0^{\frac{\pi}{2}} \frac{d\theta}{2 + \sin \theta}$$
3

(c) (i) Split into partial fractions: $\frac{8}{(x+2)(x^2+4)}$ 2

(ii) Hence evaluate: $\int_0^2 \frac{8 \, dx}{(x+2)(x^2+4)}$ 3

(d) If $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$, ($n \geq 2$)

(i) Show that $I_n = (n-1) I_{n-2} - (n-1) I_n$ 2

(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \cos^6 x \, dx$ 2

Question 2 (15 marks) Use a new 8 page booklet

Marks

- (a) If $z = 3 + 2i$, plot on an Argand diagram
- | | |
|-----------------------|---|
| (i) z and $\bar{z}$ | 1 |
| (ii) iz | 1 |
| (iii) $z(1 + i)$ | 1 |
- (b) (i) Find all pairs of integers a and b such that $(a + ib)^2 = 8 + 6i$ 1
- (ii) Hence solve: $z^2 + 2z(1 + 2i) - (11 + 2i) = 0$ 2
- (c) (i) If $z = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$, find z^6 2
- (ii) Plot on an argand diagram, all complex numbers that are the solutions of $z^6 = 1$ 2
- (d) Sketch the locus of the following. Draw separate diagrams.
- | | |
|--|---|
| (i) $\arg(z - 1 - 2i) = \frac{\pi}{4}$ | 1 |
| (ii) $\overline{zz} - 3(z + \bar{z}) \leq 0$ | 2 |
| (iii) $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{3}$ | 2 |

Question 3 (15 marks) Use a new 8 page booklet

Marks

- (a) For the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$
- (i) Find the eccentricity. 1
 - (ii) Find the coordinates of the foci S and S'. 1
 - (iii) Find the equations of the directrices. 1
 - (iv) Sketch the curve $\frac{x^2}{25} + \frac{y^2}{16} = 1$ 1
 - (v) Show that the coordinates of any point P can be represented by $(5\cos \theta, 4\sin \theta)$ 2
 - (vi) Show that $PS + PS'$ is independent of the position of P on the curve. 2
 - (vii) Show that the equation of the normal at the point P on the ellipse is $5x\sin \theta - 4y\cos \theta - 9\sin \theta \cos \theta = 0$ 2
 - (viii) If the normal meets the major axis at L and the minor axis at M, prove that $\frac{PL}{PM} = \frac{16}{25}$ 2
 - (ix) Show that the normal bisects $\angle SPS'$ 3

Question 4 (15 marks) Use a new 8 page booklet

Marks

(a) (i) Find $\int \frac{\sin 2x}{2 + \sin^2 x} dx$ **2**

(ii) Evaluate $\int_0^{\frac{1}{2}} \sqrt{\frac{1+x}{1-x}} dx$ **4**

(b) If $a > 0$, $b > 0$ and $c > 0$, show that

(i) $a^2 + b^2 + c^2 - ab - bc - ca \geq 0$ **2**

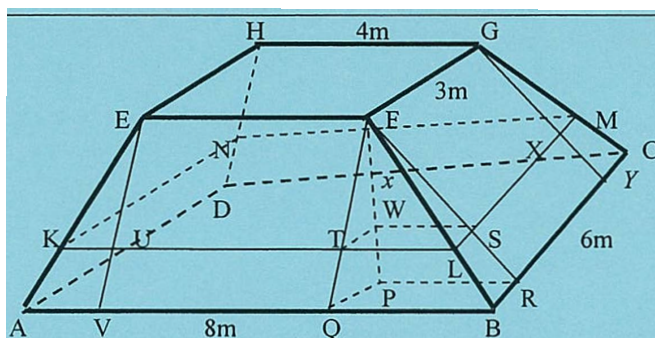
(ii) $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$ **4**

(iii) $(a+b+d)(b+c+d)(c+a+d)(a+b+c) \geq 8abcd$ **3**

Question 5 (15 marks) Use a new 8 page booklet

Marks

- (a) A concrete beam of height $15m$ has plane sides. Cross-sections parallel to the ends are rectangular. The beam measures $4m$ by $3m$ at one end and $8m$ by $6m$ at the other end as shown.



In the figure KLMN is the cross-section and its distance from the top is x metres. $FW = x$ metres

- (i) Show that an expression for the area of a cross-section at a distance x metres from the smaller end is given by $A(x) = 12 + \frac{24x}{15} + \frac{4x^2}{75}$. 3
- (ii) Find the volume of the beam. 2
- (b) Find the exact volume of the solid generated by rotating the area bounded by the curve $y = \log_e x$, the x -axis and the line $x = 4$ about the y -axis. Use the method of cylindrical shells and include sketches with your answer. 5
- (c) By slicing perpendicular to the y -axis, determine the volume formed when the region bounded by the curve $y = -3x^4 + 12x^2$ and the x -axis between $x = 0$ and $x = 2$ is rotated about the y -axis. Include sketches with your answer. 5

Question 6 (15 marks) Use a new 8 page booklet

Marks

- (a) A wasp after leaving its hive O, flies 2km North East, 4km 30° West of North and then 6km 210° True North.
- (i) Convert each of the wasp's flights into the form $z = r \operatorname{cis} \theta$. 1
- (ii) Draw a vector diagram showing the wasp's flights relative to its hive O. 1
- (iii) Determine the resultant vector of the wasp's flights. 1
- (iv) Hence determine how far to the nearest 0.1km and in what direction in radians to 1 decimal place is the wasp from its hive. 2
- (b) (i) Given that $P(x)$ has a rational zero, find this zero and hence factorise $P(x)$ over the complex field of numbers if $P(x) = 2x^3 - 3x^2 + 2x - 3$. 2
- (ii) If α, β and γ are the roots of the equation $x^3 + qx^2 + r = 0$, where $r \neq 0$ determine a cubic equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$. 2
- (iii) The equation $2x^3 - 13x^2 - 26x + 16 = 0$ has roots in geometric progression. Find these roots. 2
- (c) The tangent at $P(a \sec \theta, b \tan \theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets a directrix at Q. S is the corresponding focus.
- (i) Find the equation of the tangent at P. 1
- (ii) Find the coordinates of Q. 1
- (iii) Show that PQ subtends a right angle at S. 2

Question 7 (15 marks) Use a new 8 page booklet

Marks

- (a) Given $y = \frac{x^3}{x^2 - 4}$
- (i) Find the coordinates of all stationary points. 2
- (ii) Find the points of intersection with the coordinate axes and the position of all asymptotes. 1
- (iii) Hence sketch the curve $y = \frac{x^3}{x^2 - 4}$ 1
- (b) Use the graph $y = \frac{x^3}{x^2 - 4}$ to find the number of roots of the equation $x^3 - k(x^2 - 4) = 0$ for varying value of k . 2
- (c) (i) Show that $\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$ 2
- (ii) Hence show that $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{1 + \sin x} dx = \int_0^{\frac{\pi}{4}} \frac{2}{\cos^2 x} dx$ 2
- (iii) Hence evaluate $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{1 + \sin x} dx$ 1
- (d) A sociologist believes that the fraction $y(t)$ of a population who have heard a rumour after t days can be modelled by a continuous function given by $y(t) = \frac{y_0 e^{kt}}{(1 - y_0) + y_0 e^{kt}}$, $t \geq 0$, where y_0 is the fraction, $0 \leq y_0 < 1$ for all $t \geq 0$, who have heard the rumour at time $t = 0$ and k is a positive constant.
- (i) Show that $y_0 \leq y(t) < 1$ for all $t \geq 0$. 1
- (ii) Find the rate of change of y with respect to t . 1

(iii) If $k = 0.2$ and $y_0 = 0.1$, show that $y(5) = \frac{e}{e+9}$ **1**

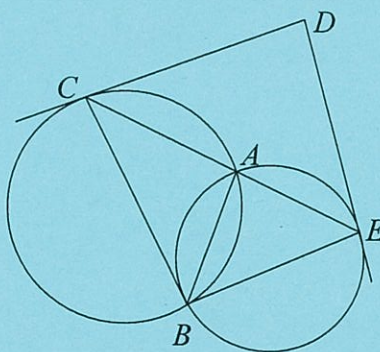
(iv) Give an interpretation of the above results (i), (ii) and (iii) in terms of the sociological model. **1**

Question 8 (15 marks) Use a new 8 page booklet

Marks

- (a) (i) Find a polynomial $p(x)$ with real coefficients having $3i$ and $1+2i$ as zeros. 2
- (ii) Find all zeros of the equation $6x^4 - 7x^3 - 28x^2 + 35x - 10 = 0$. 3
- (b) (i) If k is a positive integer such that $k \geq 4$, show that $2k^3 > 3k^2 + 3k + 1$. 2
- (ii) Hence show by mathematical induction for positive integers n , $n \neq 3$, that $3^n > n^3$. 4

(c)



Two circles intersect at A and B . CAE is a straight line where C is a point on the first circle and E is a point on the second circle. The tangent at C to the first circle and the tangent at E to the second circle meet at D .

- (i) Copy the diagram.
- (ii) Prove that $BCDE$ is a cyclic quadrilateral. 4

Q.1. (a) $\sqrt{5^2+2^2} = \sqrt{29}$ (1)

(16) (b) $\frac{1}{(-3-4i)} \times \frac{-3+4i}{(-3+4i)} = \frac{-3+4i}{9+16} = \underline{\underline{-\frac{3}{25} + \frac{4}{25}i}}$ (2)

(c) $\frac{1+i^5}{1-i} = \frac{1+i}{1-i} \times \frac{1+i}{1+i} = \frac{(1+i)^2}{1+1} = \frac{1+2i+i^2}{2} = \frac{2i}{2} = \underline{\underline{i}}$ (2)

(d) (i) $|-1+i| = \sqrt{2}$ & $|\sqrt{3}+i| = \sqrt{4} = 2$ Hence $|z| = \frac{\sqrt{2}}{2}$

$\arg(-1+i) = \frac{3\pi}{4}$ & $\arg(\sqrt{3}+i) = \frac{\pi}{6}$ Hence $\arg(z) = \frac{7\pi}{12}$

ie. $\underline{\underline{Z = \frac{\sqrt{2}}{2} \text{cis}\left(\frac{7\pi}{12}\right)}}$ (2)

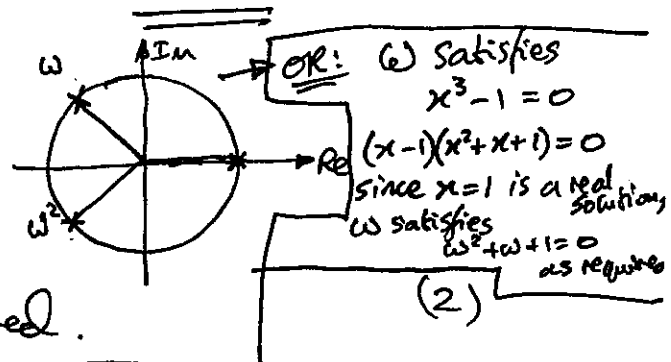
(ii) $\text{Re}(z) = \frac{\sqrt{2}}{2} \cos \frac{7\pi}{12} = \text{Re}\left(\frac{(-1+i)}{(\sqrt{3}+i)} \times \frac{(\sqrt{3}-i)}{(\sqrt{3}-i)}\right)$
 $= \text{Re}\left(\frac{-\sqrt{3}+1+i(\sqrt{3}+1)}{3+1}\right)$

$\therefore \frac{\sqrt{2}}{2} \cos \frac{7\pi}{12} = \frac{1-\sqrt{3}}{4}$

Thus $\cos \frac{7\pi}{12} = \frac{1-\sqrt{3}}{2\sqrt{2}} = \underline{\underline{\frac{\sqrt{2}-\sqrt{6}}{4}}}$ (2)

(e) (i) $\omega = \text{cis} \frac{2\pi}{3}$ or $-\text{cis} \frac{2\pi}{3}$
 then $\omega^2 = -\text{cis} \frac{2\pi}{3}$ or $\text{cis} \frac{2\pi}{3}$
 $\omega + \omega^2 = 2\cos \frac{2\pi}{3} = -1$

Hence $\omega^2 + \omega + 1 = 0$ as required.



(ii) $\text{RHS} = (b+c)(b+c\omega)(b+c\omega^2) = (b+c)(b^2+bc\omega^2+bc\omega+c^2\omega^3)$
 $= b^3+b^2c\omega^2+b^2c\omega+bc^2\omega^3+b^2c+bc^2\omega^2+bc^2\omega+c^3\omega^3$
 $= b^3+b^2c(\omega^2+\omega+1)+bc^2(\omega^3+\omega^2+\omega)+c^3\omega^3$
 $= b^3+b^2c \times 0 + bc^2(1+\omega^2+\omega) + c^3$
 Since $\omega^3=1$
 $= b^3+c^3 = \text{LHS as required.}$ (2)

(f) $\underline{\underline{P(x) = (x-2i)(x+2i)(x-1+3i)(x-1-3i) = (x^2+4)((x-1)^2+9) = (x^2+4)(x^2-2x+10)}}$
 $\underline{\underline{P(x) = x^4 - 2x^3 + 10x^2 - 8x + 40 = x^4 - 2x^3 + 14x^2 - 8x + 40}}$ (3)

2. (a) $\int \frac{x}{\sqrt{x+1}} dx$ let $u^2 = x+1$ $x = u^2 - 1$
 $2u du = dx$

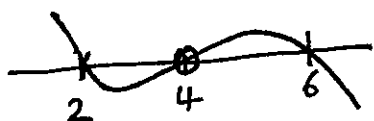
$$= \int \frac{u^2 - 1}{u} \cdot 2u du = \frac{2u^3}{3} - 2u + C = \frac{2}{3}u(u^2 - 3) + C$$

$$= \frac{2}{3}\sqrt{x+1}(x+1-3) + C = \underline{\underline{\frac{2\sqrt{x+1}(x-2)}{3} + C}} \quad (2)$$

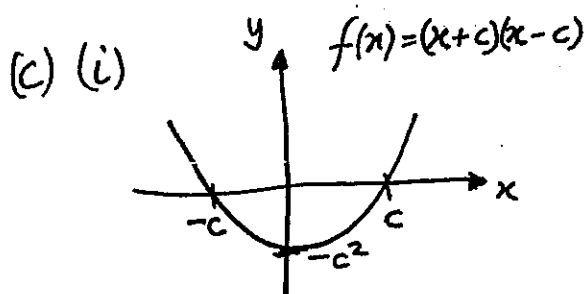
(b) $\frac{x^2 - 5x}{4 - x} + 3 \leq 0$

$$\frac{x^2 - 5x + 3(4 - x)}{4 - x} \leq 0 \quad \therefore \frac{x^2 - 8x + 12}{4 - x} \cdot (4 - x)^2 \leq 0$$

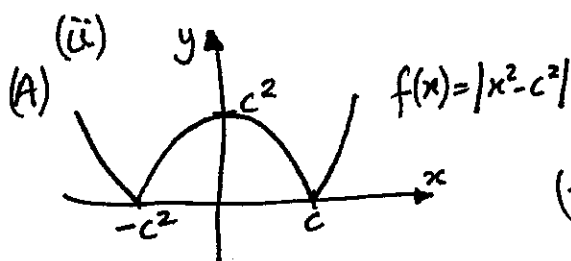
$$(x - 6)(x - 2)(4 - x) \leq 0, x \neq 4$$



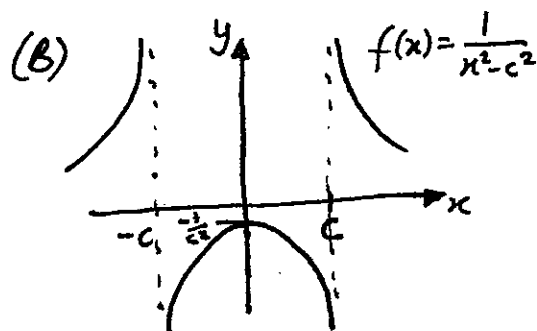
Thus $2 \leq x < 4$ or $x \geq 6$ (3)



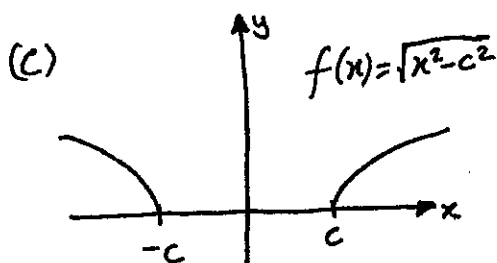
(1)



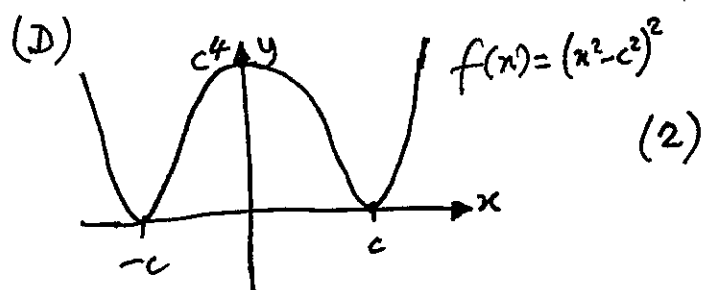
(2)



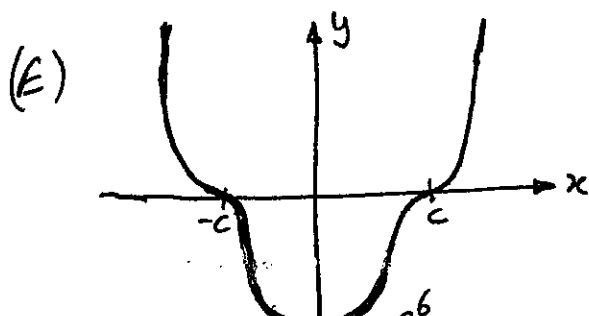
(2)



(2)



(2)



(2)

3. (a) $f(x) = x - \ln(1+x^2)$

(16) $f'(x) = 1 - \frac{2x}{1+x^2}$

$f'(x) = \frac{1+x^2-2x}{1+x^2} = \frac{(x-1)^2}{1+x^2}$

let $y = \ln u$ where $u = 1+x^2$

$\frac{dy}{du} = \frac{1}{u}$ & $\frac{du}{dx} = 2x$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{2x}{u} = \frac{2x}{1+x^2}$

Since square numbers ≥ 0 , $(x-1)^2 \geq 0$ and $1+x^2 \geq 0$ thus $f'(x) \geq 0$ for all x , as required (3)

(b) (i) In $\triangle ABX$ and $\triangle ADC$,

$\angle ABX = \angle ADC$

$\angle CAD = \angle XAB$

(Angles subtended by chord AC at the circumference in the same segment are equal)
(AD bisects angle $\angle BAC$ - given).

Therefore $\triangle ABX \parallel \triangle ADC$ (Two angles the same). (2)

(ii) Since $\triangle ABX \parallel \triangle ADC$, $\frac{AB}{AX} = \frac{AD}{AC} \therefore \underline{AB \cdot AC = AD \cdot AX}$ (1)
as required.

(iii) $\triangle CXD \parallel \triangle AXB$ since $\angle ABX = \angle ADC$ as above
and $\angle CXD = \angle BXA$ (vertically opposite angles equal).

Thus $\frac{XD}{XB} = \frac{XC}{AX} \therefore AX \cdot XD = XC \cdot XB$ so $\underline{XD = \frac{XC \cdot XB}{AX}}$ (1)

from (ii), $AB \cdot AC = AD \cdot AX = (AX + XD)AX = AX^2 + XD \cdot AX$

and from (1), $AB \cdot AC = AX^2 + \left(\frac{XC \cdot XB}{AX}\right) \cdot AX$

$AB \cdot AC = AX^2 + BX \cdot XC$ as required (2)

(c) (i) $(l+m+n)^2 = l^2 + m^2 + n^2 + 2(lm + ln + mn) = 29 + 2(lm + ln + mn)$

Thus $lm + ln + mn = \frac{(-3)^2 - 29}{2} = -10$

Let the cubic equation be monic,

ie. $x^3 + bx^2 + cx + d = 0$ where $-b = l+m+n = -3 \therefore b = 3$
 $c = lm + mn + ln = -10 \therefore c = -10$
 $-d = lmn = -6 \therefore d = 6$ (2)

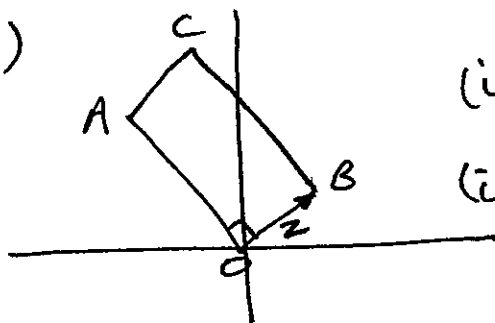
Hence monic cubic is $x^3 + 3x^2 - 10x + 6 = 0$ as required.

(ii) root at $x=1$ by inspection, ie. $(x-1)(x^2 + 4x - 6) = 0$

Hence other roots at $x = \frac{-4 \pm \sqrt{16+24}}{2} = -2 \pm \sqrt{10}$

Thus l, m and n are $1, -2+\sqrt{10}, -2-\sqrt{10}$ in any order (3)

3.(d)



(i) $A = 2iz$ by inspection (1)

(ii) $\vec{OD} = \frac{1}{2} \vec{OC} = \frac{1}{2} (\vec{OB} + \vec{OA})$
 $= \frac{1}{2} (z + 2iz)$
 $= \underline{\underline{\frac{z}{2} + iz \text{ or } z(\frac{1}{2} + i)}} \quad (2)$

4.(a) let $\cos \theta = x$ then $\cos 3\theta = 4x^3 - 3x$

$$8x^3 - 6x + 1 = 0$$

$$2(4x^3 - 3x) + 1 = 0$$

$$2\cos 3\theta + 1 = 0$$

$$\cos 3\theta = -\frac{1}{2}$$

$$3\theta = \pm \frac{2\pi}{3} + 2n\pi, \quad n \in \mathbb{Z}$$

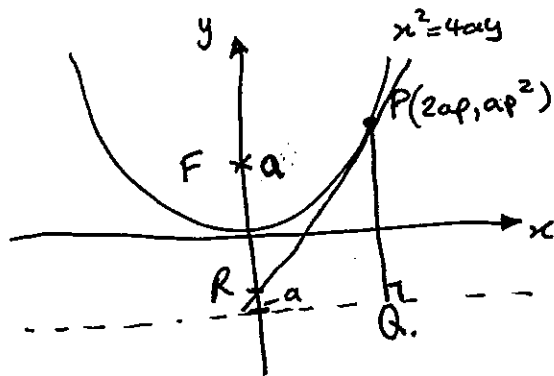
$$\therefore \theta = \pm \frac{2\pi}{9} + \frac{2n\pi}{3}, \quad n \in \mathbb{Z} \quad (3)$$

Thus $x = \underline{\underline{\cos \frac{2\pi}{9}, \cos \frac{8\pi}{9}, \cos \frac{14\pi}{9}}} \approx 0.766, -0.9397, 0.1736$
 (or equivalent!)

(b)(i) $(x-y)^2 \geq 0$ since a square ^{of a real} number is positive. (2)
 $x^2 + y^2 - 2xy \geq 0 \quad \therefore \underline{\underline{x^2 + y^2 \geq 2xy \text{ as required}}}$

BAD QUESTION! (ii) $a^2 + b^2 \geq 2ab$
 $c^2 + d^2 \geq 2cd \quad \therefore a^2 + b^2 + c^2 + d^2 \geq 2ab + 2cd$
 The question is incorrect (take $a=2, b=1, c=1, d=1$)
 $2^2 + 1^2 + 1^2 + 1^2 = 7 \neq 4 \times 2 \times 1 \times 1 \times 1$
 perhaps the question meant:
 $(a^2 + b^2)(c^2 + d^2) \geq 4abcd ? \quad (2?)$
 True since $a^2 + b^2 \geq 0$ and $c^2 + d^2 \geq 0$
 so $a^2 + b^2 + c^2 + d^2 \geq 0$

4. (c)



(i) $\frac{dy}{dx} = \frac{dy}{dp} \times \frac{dp}{dx} = 2ap \times \frac{1}{2a} = p$ $\therefore$ Eqn of tangent at P is:

$$y - ap^2 = p(x - 2ap)$$

$$y = px - 2ap^2 + ap^2$$

$$\therefore \underline{px - y - ap^2 = 0 \text{ as required.}} \quad (2)$$

(ii) Midpoint of PR is $x = ap$ and $y = p(ap) - ap^2$ from above
ie. $(ap, 0)$

Midpoint of FQ is $(ap, \frac{a - a}{2})$ or $(ap, 0)$

Hence since midpoints coincide, they bisect each other. as required (3)

(iii) Gradient of PR = $m_{PR} = p$ from (i)

Gradient of FQ = $m_{FQ} = \frac{-2a}{2ap} = -\frac{1}{p}$

$m_{PR} \times m_{FQ} = -1$ (2)

$\therefore$ perpendicular as required

(iv) PQRF is a rhombus since the diagonals bisect at right angles. (2)

5. (a) (i) $y = uv$ and $y' = uv' + u'v$ from the product rule. (2)

$y'' = uv'' + u'v' + u'v' + u''v = \underline{uv'' + 2u'v' + u''v}$ as required.

(ii) $y''' = u'v'' + uv''' + 2u''v' + 2u'v'' + u'''v + u''v'$

$y''' = \underline{uv''' + 3u''v' + 3u'v'' + u'''v}$

Similarly $y^{iv} = uv^{iv} + 4u'v''' + 6u''v'' + 4u'''v' + u^{iv}v$

and $y^v = uv^v + 5u'v^{iv} + 10u''v''' + 10u'''v'' + 5u^{iv}v' + u^v v$ (2)

BORING QUESTIONS!

(iii) $\frac{d^5}{dx^5}((1-x^2)e^{-x}) = (1-x^2)(-e^{-x}) + 5(2x)e^{-x} + 10(2)(-e^{-x})$

$= e^{-x}(x^2 - 1 + 10x - 20) = \underline{(x^2 + 10x - 21)e^{-x}}$ (2)

Q.5. (b) (i) $t_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n-1} + \frac{1}{2n}$

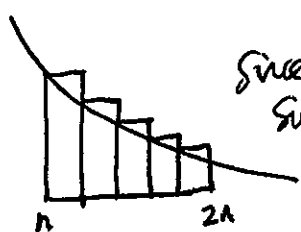
$\therefore t_n + \frac{1}{2n} = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n-1} + \frac{2}{2n}$ and $\frac{2}{2n} = \frac{1}{n}$

$t_n + \frac{1}{2n} = \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n-1}$ as required. (1)

(ii) Area under graph from $x=n$ to $2n = \int_n^{2n} \frac{1}{x} dx = \ln\left(\frac{2n}{n}\right) = \ln 2$

and area of the rectangles (width 1)

from n to $2n-1$ is $\frac{1}{n} + \frac{1}{n+1} + \dots + \frac{1}{2n-1} = t_n + \frac{1}{2n}$



Since the area formed from summing the rectangles > area under graph, (3)

then $t_n + \frac{1}{2n} > \ln 2$ as required.

(iii) for $n=1$, $S_n = 1 - \frac{1}{2} = \frac{1}{2}$ and $t_n = \frac{1}{2} \therefore$ true for $n=1$

Assume true for $n=k$, i.e. $S_k = t_k$

then for $n=k+1$, $S_{k+1} = S_k + \frac{1}{2k+1} - \frac{1}{2k+2}$ ----- (1)

$t_{k+1} = \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k+1} + \frac{1}{2k+2} = t_k + \frac{1}{2k+1} + \frac{1}{2k+2} - \frac{1}{k+1}$

Hence $t_{k+1} = S_k + \frac{1}{2k+1} + \frac{1-2}{(2k+2)}$ (4)

$= S_k + \frac{1}{2k+1} - \frac{1}{2k+2} = S_{k+1}$ from (1)

Hence true for S_{k+1}

Therefore, true for $n=1$ and if true for $n=k$, then also true for $n=k+1$
 $\therefore$ true for all $n \in \mathbb{Z}, n \geq 1$ by induction.

(iv) $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{9999} - \frac{1}{10000} = S_{5000} = t_{5000}$

As $n \rightarrow \infty$, $t_n + \frac{1}{2n} \rightarrow \ln 2$ or $t_n \rightarrow \ln 2 - \frac{1}{2n}$

Hence $t_{5000} \approx \ln 2 - \frac{1}{10000} \approx \underline{\underline{0.693}}$ (3.d.p.) (2)

$$S_1 = 1 - \frac{1}{2} = \frac{1}{2}$$

$$S_2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{7}{12}$$

$$S_3 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} = \frac{37}{60}$$

$$t_1 = \frac{1}{2}$$

$$t_2 = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

$$t_3 = \frac{1}{4} + \frac{1}{5} + \frac{1}{6} = \frac{37}{60}$$

S23

$$\text{RTP } \sum_{r=1}^n \frac{r}{(r+1)!} = \frac{(n+1) - 1!}{(n+1)!}$$

Prove true for $n=1$

$$\text{LHS} = \frac{1}{(1+1)!}$$

$$= \frac{1}{2}$$

$$\text{RHS} = \frac{1+1 - 1!}{1+1}$$

$$= \frac{1}{2}$$

~~$$\text{LHS} = \frac{1}{(3+1)!}$$~~

$$\text{LHS} = \left(\frac{1}{2} + \frac{1}{3} \right) + \frac{1}{8}$$

$$= \frac{21}{24}$$

$$= \frac{7}{8}$$

$$\text{RHS} = \frac{23}{24}$$

 $9^n - 7^n$ is evenProve true for $n=1$

$$\text{LHS} = 9 - 7$$

$$= 2$$

 $\therefore$ true for $n=1$ assume true for $n=k$

$$\text{i.e. } 9^k - 7^k = 2M \quad M \in \mathbb{Z}^+$$

prove true for $n=k+1$

$$9^{k+1} - 7^{k+1} = 2N \quad N \in \mathbb{Z}^+$$

$$\text{LHS}_2 \quad 9^{k+1} - 7^{k+1}$$

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